CLASS 9 — SCIENCE · FORMULA SHEET

Chapter 3: **The Atomic Foundations of Matter** | Quick revision, one page

Laws of chemical combination

- Law of conservation of mass (Lavoisier): total mass of reactants = total mass of products.
- Law of constant proportions (Proust): a pure compound always contains the same elements in the same fixed mass ratio. In water $\mathrm{H} : \mathrm{O} = 1 : 8$ by mass; in $\mathrm{CO_2}$ $\mathrm{C} : \mathrm{O} = 3 : 8$.
- Dalton: matter is made of indivisible atoms; atoms of one element are identical; atoms combine in simple whole-number ratios; atoms are neither created nor destroyed in a reaction.

Atoms, ions and molecules

- Atomic mass unit: $1\ \mathrm{u} = \dfrac{1}{12}$ of the mass of one carbon-12 atom.
- Molecule: two or more atoms bonded together (e.g. $\mathrm{O_2}$, $\mathrm{H_2O}$). Atomicity = number of atoms in a molecule: He 1, $\mathrm{O_2}$ 2, $\mathrm{O_3}$ 3, $\mathrm{P_4}$ 4, $\mathrm{S_8}$ 8.
- Ion: charged atom or group of atoms. Cation (positive), anion (negative); examples: $\mathrm{Na^+}$, $\mathrm{Ca^{2+}}$, $\mathrm{Cl^-}$, $\mathrm{SO_4{}^{2-}}$, $\mathrm{OH^-}$, $\mathrm{NH_4{}^+}$.

Writing chemical formulae

- Write symbols with the cation first, write valencies, then criss-cross them and reduce to the simplest ratio.
- Examples: $\mathrm{Al^{3+}}$ and $\mathrm{O^{2-}} \rightarrow \mathrm{Al_2O_3}$; $\mathrm{Ca^{2+}}$ and $\mathrm{OH^-} \rightarrow \mathrm{Ca(OH)_2}$; $\mathrm{Na^+}$ and $\mathrm{SO_4{}^{2-}} \rightarrow \mathrm{Na_2SO_4}$.
- Use brackets for polyatomic ions when more than one is needed, e.g. $\mathrm{(NH_4)_2SO_4}$.

Masses

- Molecular mass = sum of the atomic masses of all atoms in the molecule: $\mathrm{H_2O} = 18$ u, $\mathrm{CO_2} = 44$ u, $\mathrm{O_2} = 32$ u.
- Formula unit mass (ionic compounds) = sum of atomic masses in the formula unit: $\mathrm{NaCl} = 58.5$ u, $\mathrm{CaCO_3} = 100$ u, $\mathrm{Al_2O_3} = 102$ u.
- Mass per cent of an element $= \dfrac{\text{mass of the element in one formula}}{\text{molecular or formula mass}} \times 100$.

Ionic and molecular compounds

- Ionic compounds (e.g. NaCl): crystalline solids, high melting points, conduct when molten or dissolved.
- Molecular compounds (e.g. $\mathrm{CCl_4}$, HCl): low melting and boiling points, mostly gases or liquids.

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